LHCb Open Data Analysis

A CERN Open Data Analysis of Three-Body B-Meson Decays

Invariant-Mass Fits, Raw Charge Asymmetry, and Dalitz-Plot Diagnostics

Yanis Idir

Abstract

This report presents a compact but realistic high-energy-physics analysis workflow based on LHCb CERN Open Data. Three-body B-meson candidates are reconstructed from charged-hadron tracks under a kaon mass hypothesis, selected with transparent quality and particle-identification requirements, and their invariant-mass spectrum is fitted by extended binned maximum likelihood. Fitting the B^+ and B^- spectra separately gives a raw charge asymmetry of $-0.0353 \pm 0.0290\,(\text{stat.}) \pm 0.0323\,(\text{syst.})$, where the systematic uncertainty covers the choice of mass-fit model and fitted mass range under pre-registered acceptance criteria, and where the statistical procedure is validated on 2000 pseudo-experiments. The workflow is completed by a sideband-labelled classifier study, Dalitz-plot distributions, and a bin-by-bin local asymmetry diagnostic across the three-body phase space. The asymmetry is raw: no detector, production, trigger or efficiency corrections are applied, and it is presented as a diagnostic of the analysis chain rather than as a measurement of CP violation. The report emphasizes reproducibility, modular scientific code, and explicit statements of what each result does and does not establish.

October 26, 2026

Contents

1 Introduction **1**

2 The LHCb Experiment **1**

3 Dataset and Software Framework **2**
3.1 ROOT input files 2
3.2 Python analysis workflow 2

4 Invariant Mass Reconstruction **3**
4.1 Relativistic mass formula 3
4.2 Mass spectrum before and after selection 4
4.3 Signal-plus-background fit 5

5 Event Selection **6**

6 Signal and Background Classification **7**
6.1 Sideband-labelled samples 7
6.2 Random Forest classifier 7

7 Raw Charge Asymmetry **9**
7.1 Charge-separated invariant-mass fits 9
7.2 Raw asymmetry extraction 11
7.3 Toy-MC validation 12
7.4 Systematic uncertainties 12

8 Dalitz Plot Analysis **16**
8.1 Charge-ordered Dalitz variables 16
8.2 Dalitz distribution 16
8.3 Local raw-asymmetry diagnostic 17

9 Summary and Conclusion **20**

1 Introduction

The Standard Model of particle physics describes the known elementary particles and their interactions with remarkable precision. Nevertheless, several observed features of the Universe remain only partially understood. One of the most important is the matter–antimatter asymmetry: cosmological observations indicate that the visible Universe is dominated by matter, while the known laws of particle physics contain only limited sources of charge-parity (CP) violation. In the Standard Model, CP violation arises from a single complex phase in the quark-mixing matrix [1, 2], and the size of the resulting effects is far too small to account for the observed baryon asymmetry. Precision studies of flavour-changing processes are therefore a central part of modern particle physics.

Flavour physics studies transitions between different quark flavours. In the heavy-flavour sector, hadrons containing beauty quarks provide especially rich laboratories. The large mass of the beauty quark allows many decay modes, including multibody final states in which resonant and non-resonant amplitudes can interfere. Such interference can generate local CP-asymmetric effects in restricted regions of phase space, even when the integrated asymmetry is small.

B mesons are bound states containing a beauty quark or antiquark and a lighter valence quark. Charged B mesons, denoted B^+ and B^-, form a natural pair of charge-conjugate systems. By comparing the observed yields or distributions of B^+ and B^- candidates, one can construct raw asymmetry observables. In a simplified counting analysis, the integrated asymmetry is written as

$$A_{\mathrm{CP}}^{\mathrm{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}, \tag{1}$$

where $N(B^+)$ and $N(B^-)$ are observed candidate counts. A full physics measurement must correct this raw quantity for background contributions, detector-induced charge asymmetries, production asymmetries, trigger effects, and selection efficiencies. The present project treats Eq. 1 as an analysis diagnostic rather than as a final CP-violation measurement.

This report documents a reproducible analysis workflow built with CERN Open Data [3]. The project focuses on a sample of reconstructed LHCb three-hadron B candidates stored in ROOT files. The analysis reconstructs the B-candidate invariant mass, applies simple physics-motivated selections, fits the mass peak, trains a signal/background classifier using sideband labels, separates charge-conjugate candidates, and produces Dalitz-plot distributions. The goal is to demonstrate a modern high-energy-physics (HEP) workflow using transparent scientific reasoning and modular Python code.

The analysis is intentionally compact. It is not presented as a publishable LHCb measurement, but as a polished student-level project that mirrors the structure of a real experimental analysis: data access, reconstruction, selection, modelling, validation plots, and interpretation of limitations. This makes it suitable as a technical portfolio project for applications in experimental particle physics.

2 The LHCb Experiment

The Large Hadron Collider beauty experiment (LHCb) is a forward spectrometer at the CERN Large Hadron Collider designed for precision studies of hadrons containing beauty and charm quarks [4]. Proton–proton collisions at the LHC produce heavy quarks predominantly in the forward and backward regions. LHCb exploits this production topology by covering a forward angular region, providing excellent tracking, vertexing, momentum resolution, and particle identification.

The detector design is particularly well suited to heavy-flavour analyses. B hadrons have lifetimes long enough to travel measurable distances before decaying. This displaced-decay topology is identified by reconstructing primary vertices from the proton–proton interaction and secondary vertices from the B-hadron decay products. The distance between these vertices and the quality

of the reconstructed secondary vertex are powerful handles for suppressing prompt combinatorial background.

Charged-particle tracking is performed with a system of silicon and tracking stations around a dipole magnet. The curvature of charged-particle trajectories in the magnetic field provides a momentum measurement and determines the charge sign. In the present analysis, the track momentum components are used directly to build four-vectors for the three daughter hadrons. The charge information is used later to classify candidates as B^+ or B^-.

Particle identification (PID) is another central ingredient. LHCb uses ring-imaging Cherenkov detectors to distinguish kaons, pions, and protons over a broad momentum range. The open-data ntuples used in this project contain per-track PID-related variables such as kaon and pion probability-like observables. A simple kaon-PID selection is applied to build a clean demonstration sample under a $B \to KKK$ mass hypothesis.

In a complete LHCb analysis, detector effects are modelled with simulation, calibration samples, and data-driven corrections. This project keeps the detector treatment deliberately simple, but it preserves the main conceptual ingredients: displaced B vertices, charged-track reconstruction, particle identification, and magnet-polarity awareness.

3 Dataset and Software Framework

3.1 ROOT input files

The project uses the CERN Open Data record *Matter Antimatter Differences (B meson decays to three hadrons)* [5], released by the LHCb collaboration in 2017 under a CC0 licence. The record contains three ROOT files with reconstructed B-candidate ntuples. The main file used throughout this report is

`data/raw/B2HHH_MagnetDown.root`.

The two companion files, kept for extensions discussed in Sec. 9, are

`B2HHH_MagnetUp.root`, `PhaseSpaceSimulation.root`.

The presence of both magnet polarities is important for realistic charge-asymmetry studies, because reversing the magnetic field helps control detector-induced left–right and charge-dependent reconstruction effects. The present report focuses on the MagnetDown sample for a compact demonstration.

The data are stored in the ROOT format [6], a standard file and object format in high-energy physics. The relevant event information is contained in a tree named `DecayTree`. A ROOT `TTree` stores columnar data: each branch corresponds to one observable, such as a momentum component, charge, PID response, vertex quality, or flight distance. This format is efficient for large event samples because only the branches needed for a given analysis step need to be read.

3.2 Python analysis workflow

The analysis is implemented as a Python workflow. Reusable functions live under `src/`; exploratory notebooks under `notebooks/`; the fit entry points under `scripts/`; and the toy-MC, systematic and Dalitz studies under `studies/`. Every study writes its numerical output as JSON and CSV under `results/`, so each number quoted in this report can be traced to the file that produced it. The main software components are listed in Table 1; they are drawn from the scientific Python stack [7, 8] and from the Scikit-HEP ecosystem [9].

Table 1: Main software tools used in the analysis workflow.

Tool	Role in the analysis
ROOT	Standard HEP file format and ntuple storage
`uproot`	Reading ROOT TTrees directly in Python
`pandas`	Tabular event manipulation and selections
`numpy`	Vectorised numerical calculations
`scipy`	Special functions used in the mass models
`iminuit`	MINUIT interface for the likelihood minimisation
`matplotlib`	Publication-style figures
`scikit-learn`	Random Forest classifier and performance metrics

The branch set used throughout the analysis includes the momentum components `H1_PX`, `H1_PY`, `H1_PZ` through `H3_PZ`, the three track charges, PID variables such as `H1_ProbK`, and B-candidate quality variables such as `B_VertexChi2` and `B_FlightDistance`. A shared selection helper loads these branches, reconstructs the B-candidate mass, and applies the common baseline cuts. This avoids duplicated logic between the mass, machine-learning, CP-asymmetry, and Dalitz notebooks.

The workflow is intentionally modular. The main reusable components are the invariant-mass reconstruction module `src/reconstruction/invariant_mass.py`, the mass models and likelihood fit in `src/fitting/`, the shared selection in `src/physics/selection.py`, the Dalitz-kinematics module `src/physics/dalitz.py`, the plotting helpers in `src/plotting/mass_plots.py`, and the classifier utilities in `src/ml/signal_background.py`. This separation mirrors the structure of a maintainable HEP analysis, where physics definitions, fit models, plotting conventions and machine-learning utilities are reusable across studies rather than duplicated in each of them.

4 Invariant Mass Reconstruction

4.1 Relativistic mass formula

The key reconstructed observable is the invariant mass of the three-hadron B candidate. For a system of particles with total energy E and total momentum $\vec{p}$, the invariant mass is defined by

$$m^2c^4 = E^2 - |\vec{p}|^2c^2. \tag{2}$$

In high-energy-physics units with $c = 1$, this becomes

$$m^2 = E^2 - p_x^2 - p_y^2 - p_z^2. \tag{3}$$

For each daughter track, the analysis uses a charged-kaon mass hypothesis,

$$m_K = 493.677\,\mathrm{MeV}/c^2, \tag{4}$$

and reconstructs the energy as

$$E_i = \sqrt{p_{x,i}^2 + p_{y,i}^2 + p_{z,i}^2 + m_K^2}. \tag{5}$$

The B-candidate four-vector is then the sum of the three daughter four-vectors:

$$E_B = \sum_{i=1}^{3} E_i, \qquad \vec{p}_B = \sum_{i=1}^{3} \vec{p}_i. \tag{6}$$

Substituting these quantities into Eq. 3 gives the reconstructed B-candidate mass. The calculation is performed event-by-event with vectorised `numpy` operations.

4.2 Mass spectrum before and after selection

Figure 1 shows the reconstructed three-kaon mass spectrum before the baseline event selection. The distribution contains a broad combinatorial component as well as a visible accumulation near the expected B-meson mass region. This broad structure is typical of a preselection-level hadron-collider sample, where many random track combinations can satisfy a loose candidate definition.

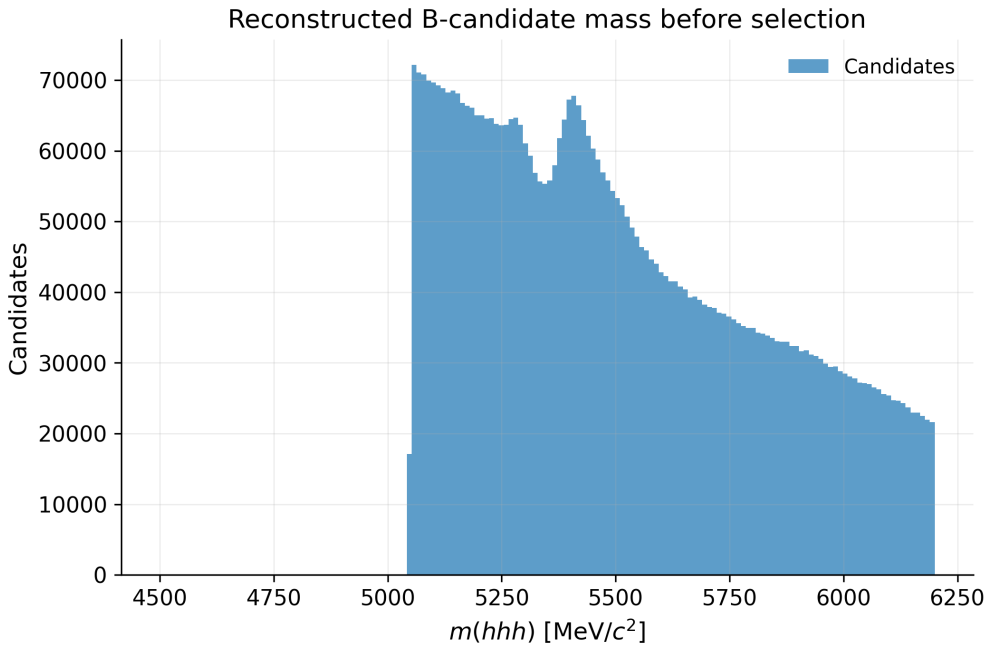


Figure 1: Reconstructed B-candidate invariant-mass distribution before baseline selection.

After applying quality, displacement, and kaon-PID selections, the mass spectrum becomes visibly cleaner, as shown in Fig. 2. The selection enhances the relative contribution of candidates compatible with a displaced B decay and suppresses a large fraction of combinatorial background.

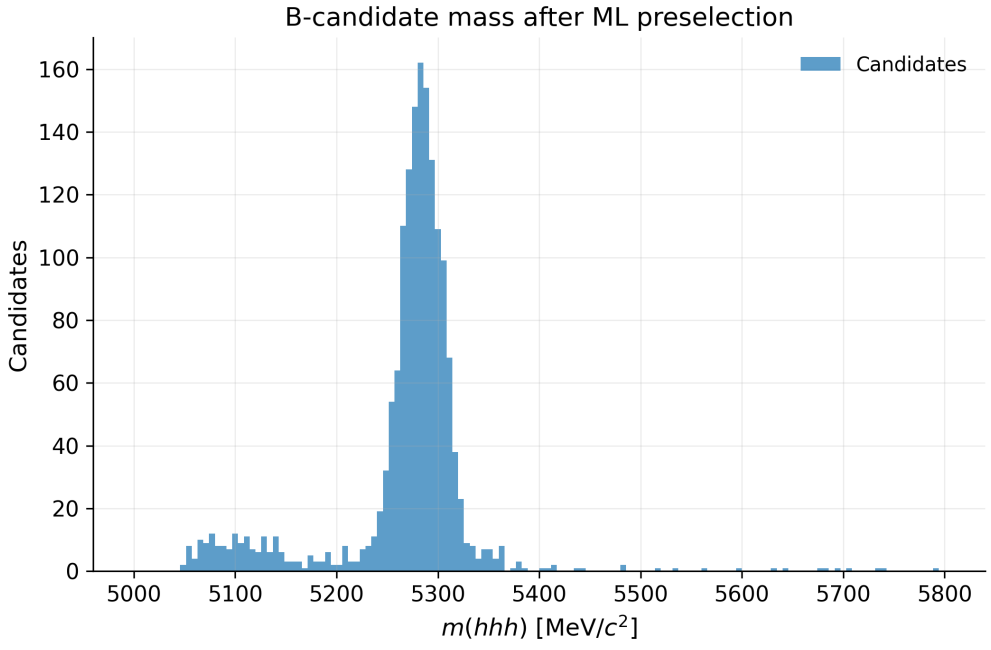


Figure 2: Reconstructed B-candidate invariant-mass distribution after baseline physics selections.

4.3 Signal-plus-background fit

A nominal signal extraction is performed with an extended binned maximum-likelihood fit of the model

$$f(m) = N_s S(m) + N_b B(m), \tag{7}$$

where $S(m)$ is the signal mass shape, $B(m)$ a finite-range exponential describing the combinatorial background, and N_s and N_b the fitted yields inside the fitted interval. The likelihood is minimised with MINUIT [10] through the `iminuit` interface.

Two choices define the nominal configuration, and both are worth stating explicitly because they are where this analysis is most fragile.

Fitted interval. The fit is performed over $5200 < m(hhh) < 5600$ MeV/c^2. The spectrum below 5200 MeV/c^2 contains a broad structure around 5050–5150 MeV/c^2, characteristic of partially reconstructed and misidentified B decays, which the combinatorial background model does not describe. Including it produced a systematic trend in the fit residuals on both charge samples. Restricting the interval removes that trend at the cost of a shorter low-mass sideband.

Signal shape. The nominal signal shape is a single Gaussian. A common-mean double Gaussian was tested as the nominal shape, since detector-level mass peaks often show a narrow core with broader tails, but on this sample its wide component, $\sigma = 52 \pm 11$ MeV/c^2, takes over the combinatorial background: the fitted background yield falls from 114 ± 17 to 22 ± 31 candidates, compatible with zero, and the uncertainty on the signal yield inflates from 39 to 66. The fit does reach a valid minimum, with a narrow-core fraction of 0.86 ± 0.04, once several starting points are tried, but with roughly 1.4×10^3 candidates and a truncated low-mass sideband the data cannot tell a broad resolution tail from the background. The double Gaussian is therefore used as a systematic variation rather than as the nominal shape.

The nominal fit is shown in Fig. 3. Of the 1645 selected candidates, 1445 fall inside the fitted interval. The results are

$$N_s = 1331 \pm 39, \tag{8}$$

$$N_b = 114 \pm 17, \tag{9}$$

$$\mu = 5284.7 \pm 0.6 \text{ MeV}/c^2, \tag{10}$$

$$\sigma = 19.5 \pm 0.5 \text{ MeV}/c^2, \tag{11}$$

with $\chi^2/\text{ndof} = 0.90$. The fitted peak position lies within a few MeV/c^2 of the known $B^\pm$ mass [11], which is the main closure check on the four-vector reconstruction and the mass hypothesis. The fitted resolution is a detector resolution convolved with the kaon mass hypothesis, not a measurement of the natural width.

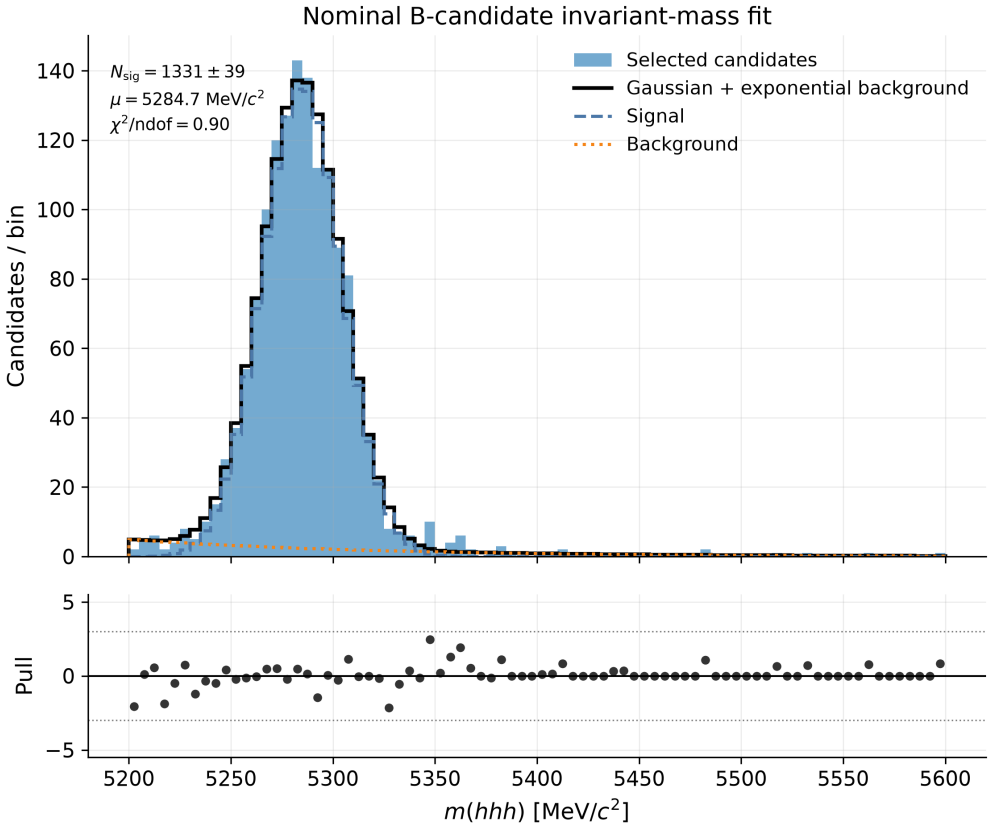


Figure 3: Selected B-candidate mass spectrum fitted with a single-Gaussian signal component and a finite-range exponential background, with fit residuals below.

5 Event Selection

Event selection is a central part of any hadron-collider analysis. The raw candidate sample contains true signal decays, partially reconstructed backgrounds, misidentified backgrounds, and random combinations of charged tracks. The goal of the baseline selection is to retain a useful signal-rich region while keeping the cuts simple enough to interpret.

The open-data ntuple is already preselected: it contains only candidates with $\chi^2_{\text{vertex}}(B) < 12$ and $d_{\text{flight}}(B) > 3$ mm. The baseline selection applied on top of that is

$$\chi^2_{\text{vertex}}(B) < 5, \tag{12}$$

$$d_{\text{flight}}(B) > 20 \text{ mm}, \tag{13}$$

$$P_K(H_1), P_K(H_2), P_K(H_3) > 0.5. \tag{14}$$

Here $\chi^2_{\text{vertex}}(B)$ is the χ^2 of the three-track vertex fit, not divided by its number of degrees of freedom; $d_{\text{flight}}(B)$ is the reconstructed distance in millimetres between the primary and secondary vertices; and $P_K(H_i)$ is the kaon PID response for daughter track H_i, a probability-like variable between 0 and 1.

The selection is severe: it keeps 1645 of the 5 135 823 candidates in the sample, a retention of 3×10^{-4}. That is a deliberate trade of efficiency for purity, and it has a consequence used later in this report: after these cuts there is little residual background left for a classifier to separate.

The vertex-quality cut suppresses candidates whose three tracks do not form a well-reconstructed secondary vertex. This is physically motivated because the daughters of a true B decay should originate from a common decay point. A poor vertex fit is more likely to arise from accidental combinations of tracks produced in different processes or from poorly reconstructed tracks.

The flight-distance selection exploits the finite lifetime of B mesons. Because B hadrons travel a measurable distance before decaying, their decay vertices are displaced from the primary proton–proton interaction point. Requiring a sufficiently large flight distance therefore suppresses prompt background combinations that originate near the primary vertex.

The PID selection implements the kaon mass hypothesis used in the invariant-mass reconstruction. Requiring all three tracks to have kaon-compatible PID values improves the consistency of the reconstructed final state. In a more complete analysis, one would optimize PID requirements with calibration samples and evaluate the corresponding efficiency and misidentification rates. Here the selection is chosen as a transparent demonstration of how detector-level PID information enters the analysis.

The selected sample is used consistently across the mass reconstruction, signal/background classification, CP-asymmetry diagnostic, and Dalitz-plot studies. This consistency is important: changing the selection between notebooks would make comparisons ambiguous and could introduce artificial differences in the resulting distributions.

6 Signal and Background Classification

6.1 Sideband-labelled samples

The signal/background study uses mass windows to create approximate labels. Candidates in a narrow window around the B mass are labelled as signal-like, while candidates in lower and upper mass sidebands are labelled as background-like. The sideband method is common in HEP because sideband regions provide a data-driven representation of combinatorial background.

The label definition used in the notebook is

$$\text{signal}: \quad 5270 < m_B < 5290 \text{ MeV}/c^2, \tag{15}$$

$$\text{background}: \quad 5200 < m_B < 5250 \text{ MeV}/c^2 \quad \text{or} \quad 5310 < m_B < 5350 \text{ MeV}/c^2. \tag{16}$$

These labels are proxy labels. The signal window can still contain background, and the sidebands may not perfectly represent the background under the signal peak. Nevertheless, the construction is useful for demonstrating a realistic machine-learning workflow without relying on simulation truth information.

6.2 Random Forest classifier

A Random Forest classifier [12] is trained using reconstruction-quality and PID variables. The B-candidate mass is deliberately excluded from the feature list, because the mass defines the labels. Including it would make the classifier learn the sideband construction rather than physically meaningful differences in candidate quality or PID response.

The classifier uses the following feature set:

$$\left\{\chi^2_{\text{vertex}}(B),\, d_{\text{flight}}(B),\, P_K(H_1), P_K(H_2), P_K(H_3),\, P_\pi(H_1), P_\pi(H_2), P_\pi(H_3)\right\}. \tag{17}$$

The label construction yields 528 signal-like and 227 background-like candidates, and the model is evaluated on a held-out third of that sample (227 candidates). The receiver operating characteristic (ROC) curve in Fig. 4 shows the trade-off between signal efficiency and background misclassification rate. The diagonal line corresponds to a classifier with no discriminating power.

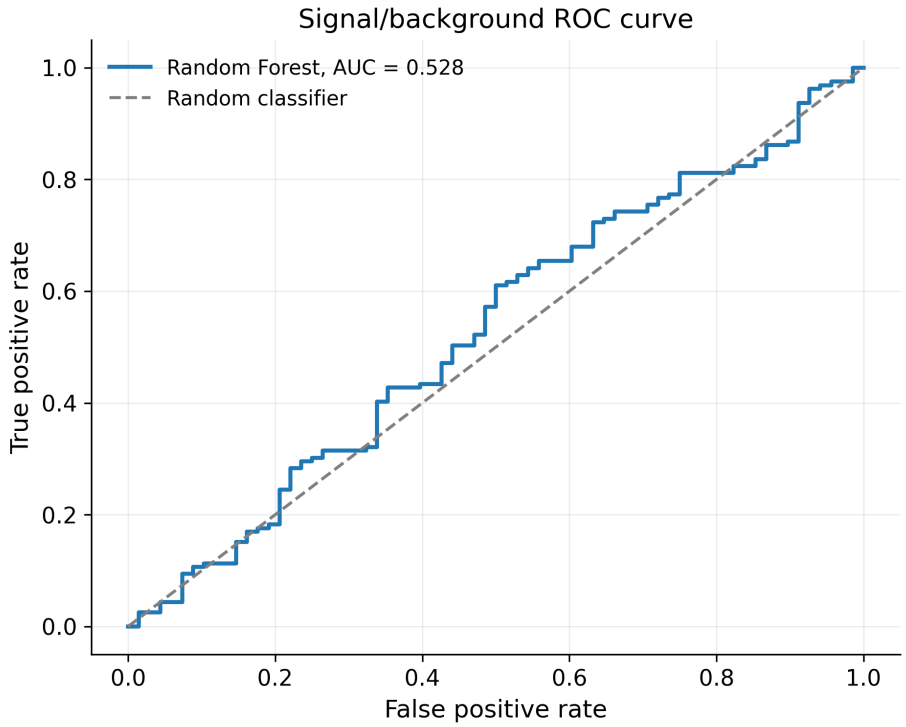


Figure 4: ROC curve for the sideband-labelled Random Forest signal/background classifier.

The measured area under the curve is AUC = 0.528. This is compatible with no separation at all, and it should be read as a negative result rather than as a performance figure. Two reasons explain it, and both are properties of the analysis rather than of the algorithm. First, the baseline selection of Sec. 5 is already tight enough that the surviving sample is dominated by signal; the classifier is asked to separate a background population that has largely been removed. Second, the features available in the ntuple are the same variables the selection already cuts on, so little independent information is left. Training before the tight preselection, on a looser sample and with additional kinematic variables, is the way to make this study informative; it is listed as a planned extension.

Figure 5 shows the feature-importance ranking from the Random Forest model. Given the AUC above, this ranking describes which variables the model used, not which variables carry physical discrimination. Such rankings depend on feature correlations, sample composition, and the chosen model architecture, and should not be read as physics observables.

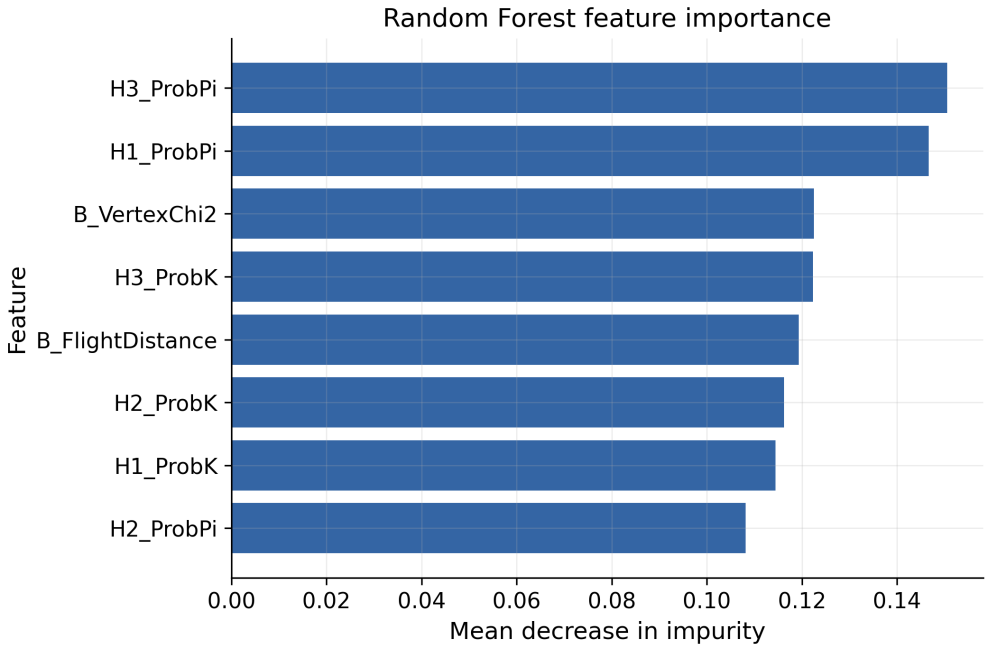


Figure 5: Random Forest feature importance for the signal/background classification task.

This part of the project demonstrates the integration of standard HEP sideband logic with modern machine-learning tools, and it also demonstrates when such a tool has nothing to add. The classifier is not used anywhere in the mass-fit, asymmetry or Dalitz results that follow; those rely on the transparent baseline selection alone. Making the classification study useful would require training before the preselection, cross-validation, gradient-boosted decision trees, systematic studies of the sideband definition, and comparisons with simulation.

7 Raw Charge Asymmetry

The selected candidates can be separated into charge-conjugate samples using the total charge of the three daughter tracks,

$$Q_{\text{tot}} = Q(H_1) + Q(H_2) + Q(H_3). \tag{18}$$

Candidates with $Q_{\text{tot}} = +1$ are classified as B^+ candidates, while candidates with $Q_{\text{tot}} = -1$ are classified as B^- candidates. This assignment uses only reconstructed track-charge information and is therefore a natural first step toward a charge-asymmetry study in a three-charged-hadron final state. The selected sample contains 884 B^+ and 761 B^- candidates.

7.1 Charge-separated invariant-mass fits

Rather than counting candidates in a fixed signal window, the asymmetry is extracted by fitting the B^+ and B^- invariant-mass spectra separately, with the nominal model of Sec. 4.3: a single Gaussian signal and a finite-range exponential background over $5200 < m(hhh) < 5600$ MeV/c^2. Fitting rather than counting matters here, because a fixed window mixes the background into the yield, and the background fraction need not be the same for the two charges.

The fitted spectra are shown in Fig. 6 and Fig. 7. Both fits converge to a valid minimum with a positive-definite covariance matrix. Goodness of fit is judged with the likelihood-ratio χ^2 of Baker and Cousins [13], which uses every bin including the empty ones. With many bins holding only a few candidates, its asymptotic χ^2 distribution does not hold — the asymptotic p-value for B^+ is 0.18 — so the p-value is taken from the 2000 pseudo-experiments of the toy study below, each refitted: $p = 0.035 \pm 0.004$ for B^+ and $p = 0.43 \pm 0.01$ for B^-. The B^+ description is

therefore marginal, at about two standard deviations. The fitted signal yields are

$$N(B^+) = 689.1 \pm 28.6, \tag{19}$$
$$N(B^-) = 642.1 \pm 26.2, \tag{20}$$

where the uncertainties are the statistical yield uncertainties returned by the binned likelihood fits.

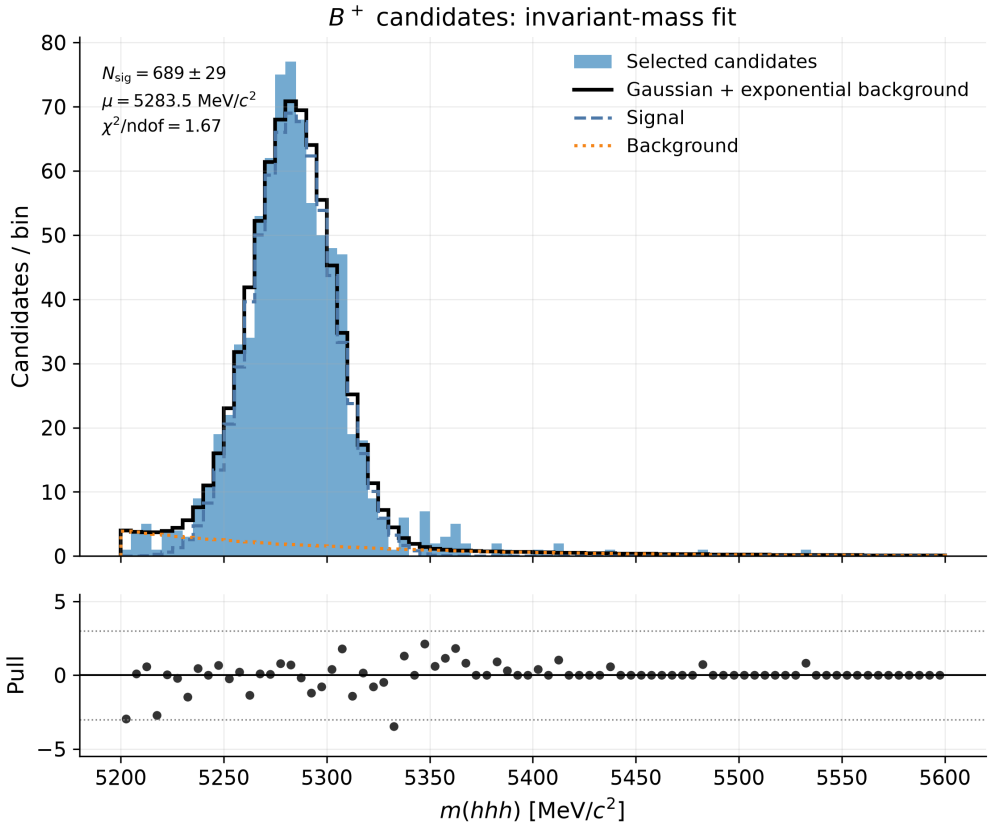


Figure 6: Charge-separated invariant-mass fit for the B^+ candidate sample.

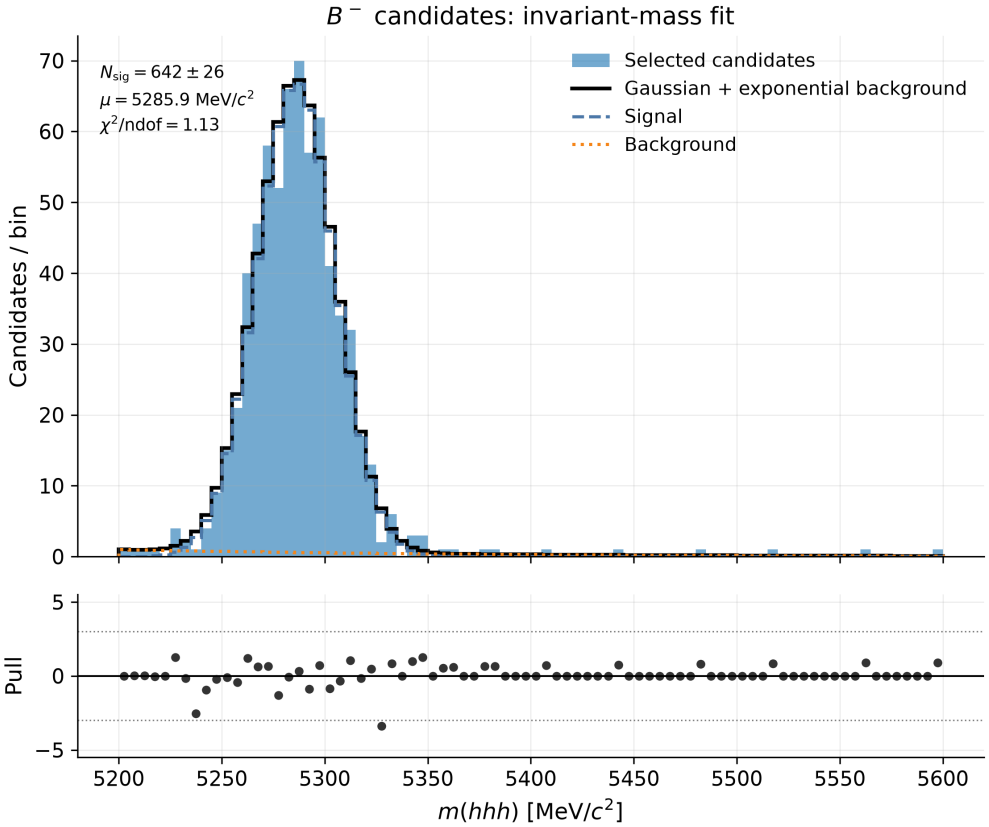


Figure 7: Charge-separated invariant-mass fit for the B^- candidate sample.

7.2 Raw asymmetry extraction

The fitted signal yields are used to construct the raw asymmetry

$$A_{\text{raw}} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}. \tag{21}$$

Propagating the independent fitted-yield uncertainties gives

$$A_{\text{raw}} = -0.0353 \pm 0.0290 \text{ (stat.)}. \tag{22}$$

After including the systematic variations described below, the result is

$$A_{\text{raw}} = -0.0353 \pm 0.0290 \text{ (stat.)} \pm 0.0437 \text{ (syst.)}, \tag{23}$$

that is, compatible with zero at the level of 0.7 standard deviations of the total uncertainty.

This is a raw asymmetry, not a physical CP-asymmetry measurement. It still contains detector-induced charge asymmetries, possible B^+/B^- production asymmetries, trigger and reconstruction effects, residual background-model dependence, and selection-induced charge biases, and it uses a single magnet polarity. A complete measurement would require detector calibration, magnet-polarity combination, background control, efficiency corrections, a control channel to cancel production and detection asymmetries, and a more complete systematic uncertainty programme. The value in Eq. 23 should therefore be interpreted as a diagnostic of the analysis workflow rather than as evidence for CP violation.

For orientation only, the published LHCb measurement of the inclusive CP asymmetry in $B^\pm \to K^\pm K^+ K^-$, using the full 2011 dataset with all corrections applied, is $A_{CP} = -0.043 \pm 0.009$ (stat.) ± 0.003 (syst.) ± 0.007 ($J/\psi K^\pm$) [14]. The raw value obtained here is of the same sign and magnitude, with an uncertainty about four times larger. That numerical proximity

is encouraging but must not be read as a validation: none of the corrections that turn a raw asymmetry into a CP asymmetry have been applied, and the uncorrected detector and production effects are not known to be small compared with the quoted uncertainty.

7.3 Toy-MC validation

The statistical machinery was validated with pseudo-experiments generated from the nominal charge-separated fitted model. In each toy experiment, B^+ and B^- mass spectra were generated with the injected fitted yields and then refitted with the same model used in the data fit. The toy study checks whether the fitting procedure and the uncertainty propagation recover the injected raw asymmetry when the fit model is correct.

A study of 2000 toys with a fixed random seed produced no fit failures and no toy rejected from the pull study. The mean raw-asymmetry pull is -0.003 and the pull width is 0.9999, both consistent with a standard normal distribution. The fitted bias is below 10^{-4}, four orders of magnitude smaller than the statistical uncertainty. The mean fitted uncertainty, 0.02889, matches the observed spread of the fitted asymmetries, 0.02888, so the uncertainty returned by the fit is correctly calibrated. This validates the statistical extraction under a correct model; it is not a detector-level systematic study, and it cannot detect a bias caused by the model itself being wrong.

7.4 Systematic uncertainties

Two analysis choices are varied: the mass-fit model and the invariant-mass fit range. A configuration is accepted only if the B^+ and B^- fits converge to a valid minimum, the fitted yields and yield uncertainties are positive, and the χ^2/ndof values remain below 2.5 for both charge-separated spectra. These criteria were fixed before the variations were run. Because the double-Gaussian model has a nearly degenerate direction, each fit is started from several points and the lowest valid minimum is kept; with a single starting point the double-Gaussian fits failed to converge for one charge and were rejected for that reason alone.

One source at a time. The fit-model study compares the nominal single Gaussian plus exponential model with a single Gaussian plus finite-range linear background and with two double-Gaussian alternatives. All three are accepted. The double-Gaussian variations shift the raw asymmetry by -0.027 and -0.031: their wide component, $\sigma \approx 53$ MeV/c^2, takes over the combinatorial background, whose fitted yield falls to values compatible with zero. The largest shift, 0.031, is the fit-model diagnostic (Fig. 8).

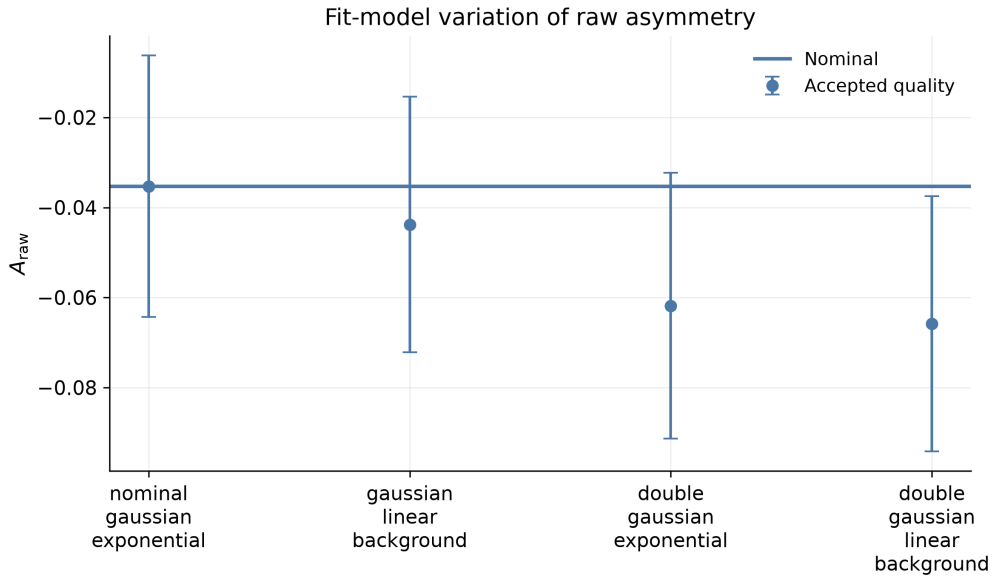


Figure 8: Raw asymmetry obtained with different charge-separated mass-fit models on the nominal fit range.

The fit-range study repeats the nominal fits for narrower and wider intervals. A change of range does not only change the analysis choice: it adds or removes candidates, and the two results would differ even with no systematic effect. The expected size of that statistical difference is $\sigma_{\mathrm{B}} = \sqrt{|\sigma^2_{\mathrm{var}} - \sigma^2_{\mathrm{nom}}|}$ (Barlow [15]), and only the part of a shift Δ beyond it, $\sqrt{\max(\Delta^2 - \sigma^2_{\mathrm{B}}, 0)}$, is evidence of a systematic effect. Raising the lower edge from 5200 to 5230 MeV/c^2 shifts the asymmetry by $\Delta = 0.031$ with $\sigma_{\mathrm{B}} = 0.015$; a toy study that fits the same pseudo-data on both ranges gives a spread of 0.0144 ± 0.0004 and no mean shift, confirming the formula. The shift is $2.1\,\sigma_{\mathrm{B}}$, a genuine effect of the data, and its Barlow-corrected size, 0.027, is the fit-range diagnostic (Fig. 9).

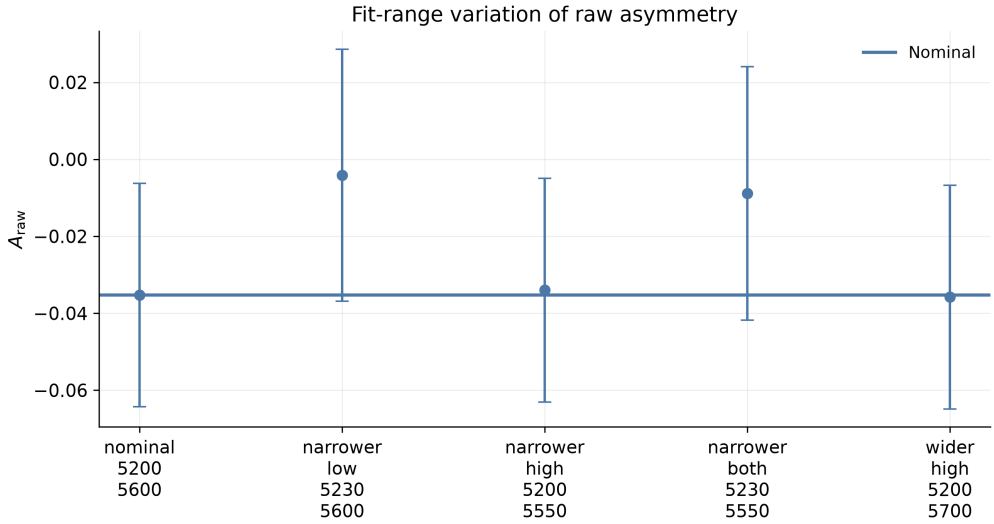


Figure 9: Raw asymmetry obtained for alternative invariant-mass fit ranges using the nominal signal and background model.

Both sources together. The two diagnostics cannot be added in quadrature, because they are not independent. Repeating every model on every range (Fig. 10) shows that the sensitivity to the lower edge belongs to the nominal model alone: with any other model, raising the edge to

5230 MeV/c^2 moves the asymmetry by at most 0.006. The nominal exponential is steep enough to absorb a shoulder on the low-mass flank of the B^+ peak that the B^- spectrum does not show; narrowing the range removes that shoulder from the fit, while the other models attribute it to the signal or to a flatter background from the start. Both diagnostics are symptoms of the same unmodelled structure.

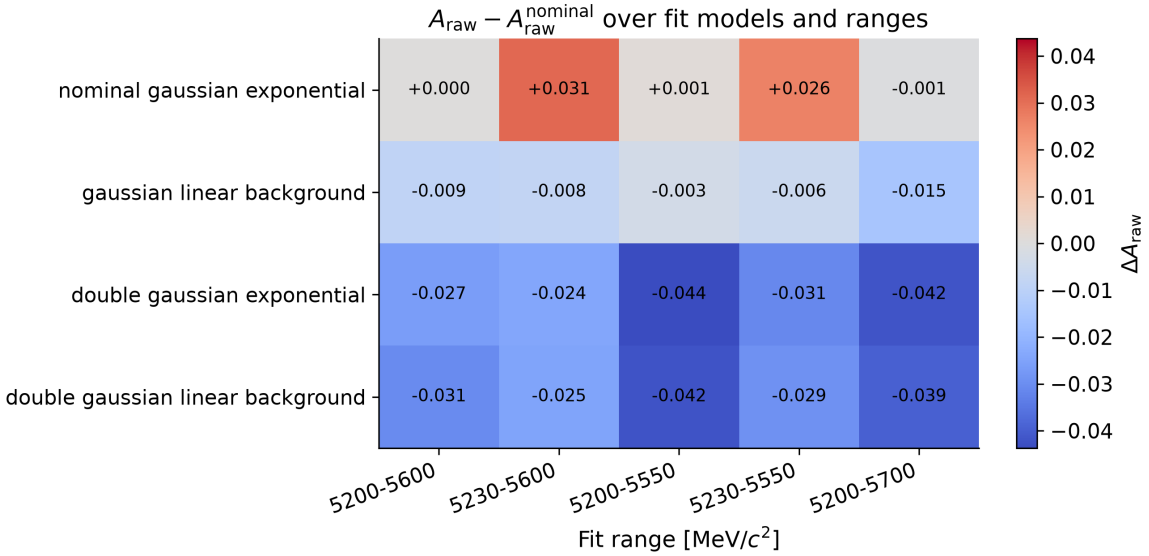


Figure 10: Shift of the raw asymmetry from the nominal for every combination of fit model and fit range. All 20 configurations pass the acceptance criteria.

The systematic uncertainty is therefore taken as the largest shift from the nominal over all 20 accepted configurations, 0.0437, reached by the double-Gaussian model on 5200–5550 MeV/c^2. This measures the joint variation directly instead of assuming a correlation. The range shifts inside it are not Barlow-corrected, so it is mildly conservative. The nominal also sits at the edge of the grid: 16 of the 19 other configurations give a more negative asymmetry.

Table 2: Uncertainty components for the fitted raw asymmetry. The one-at-a-time diagnostics are not combined; the quoted systematic is the joint envelope.

Source	Uncertainty on A_{raw}
Statistical	0.0290
Fit model alone (diagnostic)	0.0305
Fit range alone, Barlow-corrected (diagnostic)	0.0273
Systematic, joint model × range envelope	0.0437
Total	0.0525

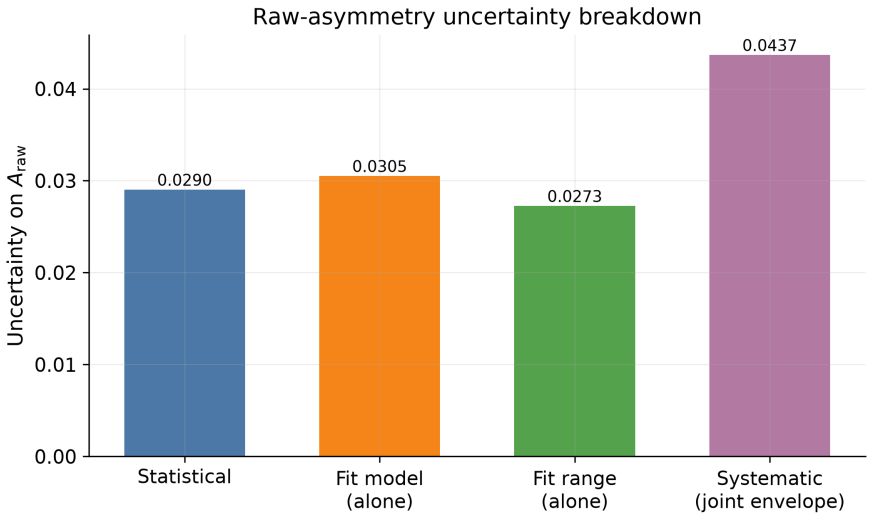


Figure 11: Statistical uncertainty, the two one-at-a-time diagnostics, and the joint systematic envelope on the raw asymmetry.

The systematic now exceeds the statistical uncertainty, and it is dominated by a single feature of the data: the low-mass structure in the B^+ spectrum, which no tested model describes. Modelling it explicitly — a partially-reconstructed or misidentified component fitted over a wider range, rather than excluded — is the first step a real measurement would need. Beyond that, signal-shape variations constrained by simulation, magnet-polarity comparisons, a control channel and detector-efficiency corrections would all be required before interpreting the raw asymmetry as a corrected CP observable.

8 Dalitz Plot Analysis

8.1 Charge-ordered Dalitz variables

Three-body decays contain more kinematic information than a single reconstructed parent mass. The internal phase-space structure is studied with pairwise invariant masses,

$$m_{ij}^2 = (E_i + E_j)^2 - |\vec{p}_i + \vec{p}_j|^2 \,, \tag{24}$$

with each daughter assigned the kaon mass hypothesis and the squared masses reported in GeV^2/c^4.

The three tracks of a candidate carry labels H_1, H_2, H_3 in the ntuple, but these labels have no physical meaning: they are not ordered by charge. A $B^+ \to K^+K^+K^-$ candidate has two opposite-sign pairs, K^+K^-, and one same-sign pair, K^+K^+. Pair masses built directly from the labels, such as $m^2(H_1H_2)$, therefore mix same-sign and opposite-sign pairs from one candidate to the next, and do not define a fixed region of phase space. They are not usable for a comparison of B^+ and B^-.

The opposite-sign pairs are instead identified from the daughter charges, as the two pairs that contain the track whose charge is opposite to the total. Because the two same-sign kaons are identical particles, the plane is folded by ordering the two opposite-sign masses,

$$m^2_{\mathrm{low}}(K^+K^-) \leq m^2_{\mathrm{high}}(K^+K^-). \tag{25}$$

These coordinates do not depend on the track labelling, and they are the ones that map onto each other under CP: the K^-K^+ pairs of the conjugate B^- decay. A B^+ and a B^- candidate in the same bin then sit in the same region of phase space, which is what a bin-by-bin asymmetry requires. Both properties are covered by unit tests.

8.2 Dalitz distribution

A Dalitz plot displays the density of selected events in the plane of two pair-mass-squared variables. Resonant two-body substructures appear as bands, while interference and phase-space effects produce non-uniform densities. In a complete amplitude analysis these structures would be modelled with coherent sums of resonant and non-resonant amplitudes; the present analysis restricts itself to visualising the distribution.

Figure 12 shows all selected candidates, without a mass window; the local asymmetry study of Sec. 8.3 restricts itself to the signal region. The narrow $\phi(1020) \to K^+K^-$ band at $m^2_{\mathrm{low}} \approx 1.04\ \mathrm{GeV}^2/c^4$ is clearly visible, which is a check that the opposite-sign pairs are correctly identified: with label-based pairs the band is diluted across the three combinations.

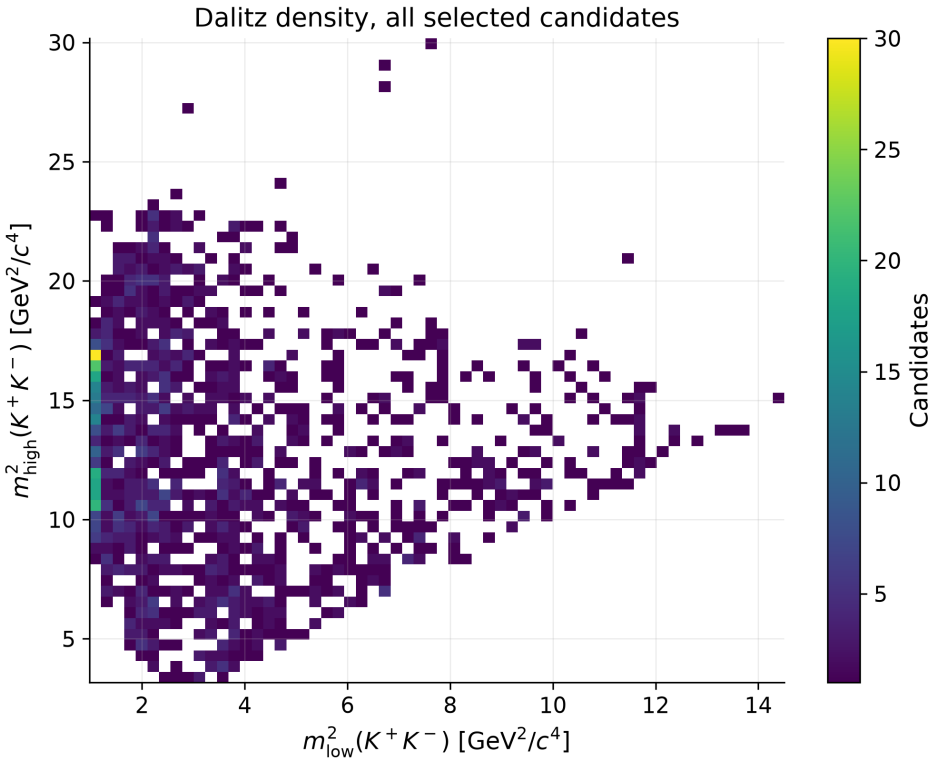


Figure 12: Dalitz density of all selected candidates in the charge-ordered $m^2_{\text{low}}(K^+K^-)$ versus $m^2_{\text{high}}(K^+K^-)$ plane.

Dalitz plots are especially relevant for CP-violation studies in multibody decays. Even if the integrated raw asymmetry is small, local asymmetries in restricted phase-space regions can be larger. This motivates a first bin-by-bin diagnostic of charge-dependent structures across the Dalitz plane.

8.3 Local raw-asymmetry diagnostic

The local diagnostic uses the candidates of the signal region, $5245 < m(hhh) < 5325$ MeV/c^2, which is the fitted peak position of Sec. 4.3 widened by twice the fitted resolution. Restricting to that window keeps the diagnostic on the same candidates the mass fits describe; using every selected candidate instead would mix in the sidebands and the low-mass structure excluded from the fits. The window contains 1323 candidates, 689 of positive and 634 of negative charge. Candidates are split into B^+ and B^- samples using the total reconstructed daughter charge,

$$q_{\text{tot}} = q_1 + q_2 + q_3. \tag{26}$$

The $m^2_{\text{low}}(K^+K^-)$ versus $m^2_{\text{high}}(K^+K^-)$ plane is then divided into regular two-dimensional bins. In each bin (i, j), the raw asymmetry is computed directly from candidate counts,

$$A^{ij}_{\text{raw}} = \frac{N^{ij}_- - N^{ij}_+}{N^{ij}_- + N^{ij}_+}, \tag{27}$$

where N^{ij}_+ and N^{ij}_- are the numbers of B^+ and B^- candidates in that Dalitz bin. The statistical uncertainty is estimated from binomial counting propagation, and a local significance is formed as $A^{ij}_{\text{raw}}/\sigma(A^{ij}_{\text{raw}})$.

The default study uses a 10×10 binning and masks bins with fewer than 20 candidates in the asymmetry and significance maps. The bin contents are still stored in the numerical output, but low-statistics regions are not interpreted visually. With this threshold, 23 out of 100 bins pass

the requirement; about half of the grid lies outside the folded kinematic region and is empty by construction. The direct counting asymmetry over the whole window is

$$A_{\mathrm{raw}} = -0.0416 \pm 0.0275, \tag{28}$$

consistent with the fit-based value of Eq. 21, as it should be: the two use the same candidates, one subtracting the fitted background and the other not. The counting value does not correct for detector efficiency, phase-space acceptance, bin migration, production asymmetry, or detector-induced charge asymmetries.

The largest local significance over the 23 populated bins is 2.3. Treating the bins as independent and each significance as a unit Gaussian under the no-asymmetry hypothesis, the probability that at least one of 23 bins fluctuates that far is

$$p_{\mathrm{global}} = 1 - (1 - p_{\mathrm{local}})^{23} = 0.40, \tag{29}$$

so this diagnostic shows no evidence of local structure — the expected outcome at this sample size. The published LHCb study of local asymmetries in this channel uses a far larger dataset and a finer binning of the Dalitz plane [14].

Figure 13 shows the selected-candidate population used to define the local bins. The occupancy is highly non-uniform, reflecting both phase-space structure and the effect of the selection. This count map is essential context for interpreting the local asymmetry and significance maps.

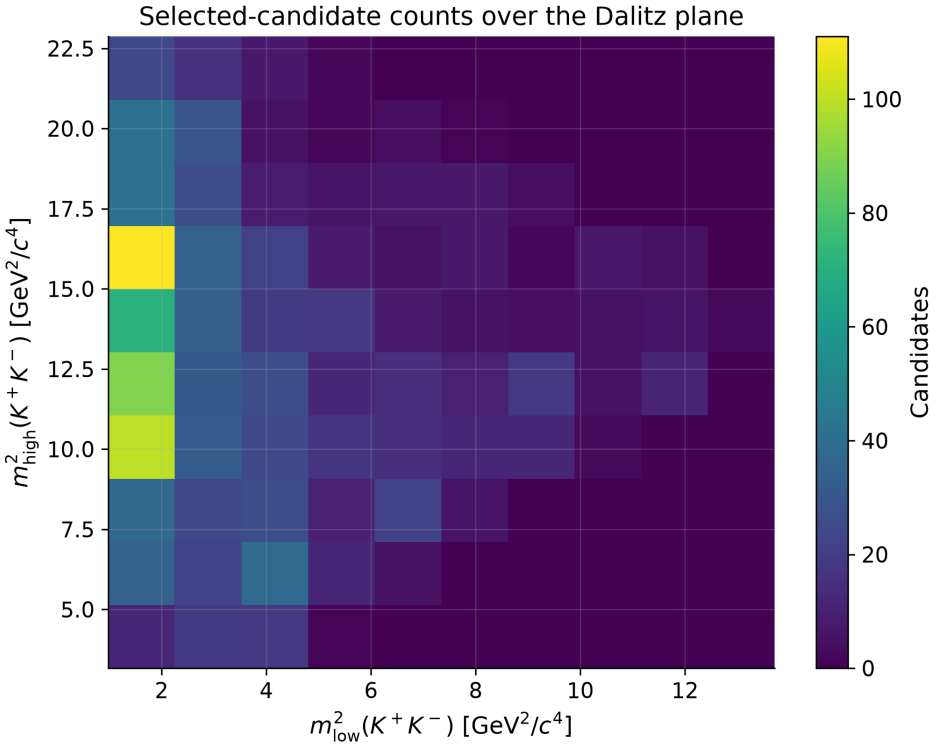


Figure 13: Selected-candidate counts in the charge-ordered $m^2_{\mathrm{low}}(K^+K^-)$ versus $m^2_{\mathrm{high}}(K^+K^-)$ Dalitz plane.

Figure 14 shows the local raw-asymmetry map after the minimum-count masking. The map is useful for visualising possible charge-dependent structures, but no physics claim is made from this diagnostic alone.

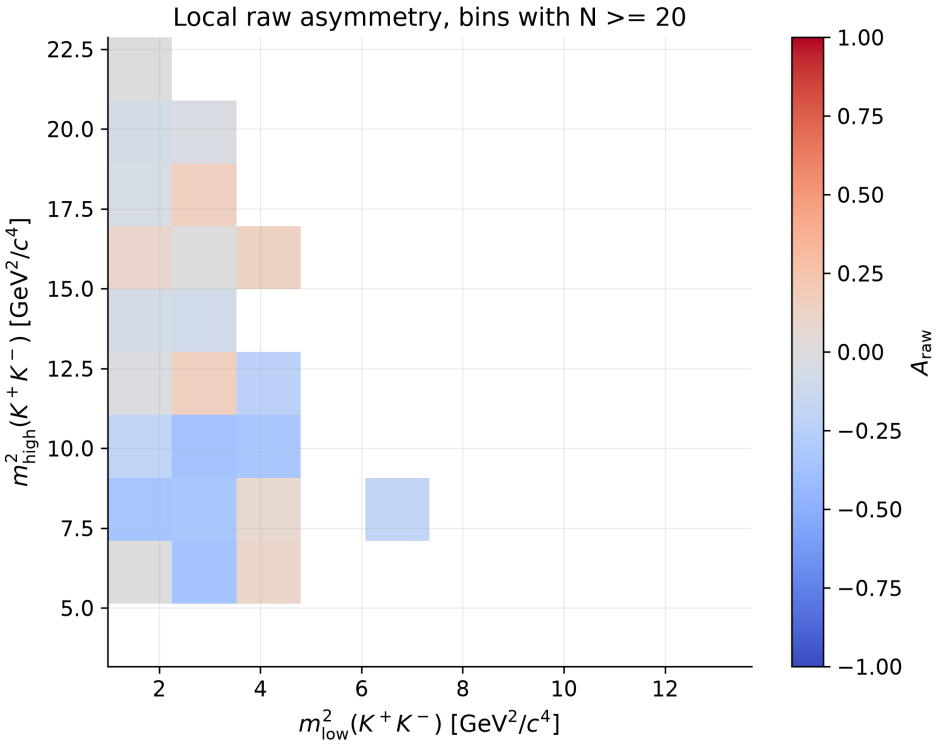


Figure 14: Local raw-asymmetry map in the charge-ordered $m^2_{\mathrm{low}}(K^+K^-)$ versus $m^2_{\mathrm{high}}(K^+K^-)$ Dalitz plane. Bins with fewer than 20 selected candidates are masked.

Figure 15 shows the corresponding local counting significance. The largest local deviations must be interpreted with caution because many bins are inspected and no look-elsewhere correction, background subtraction, or efficiency correction is applied.

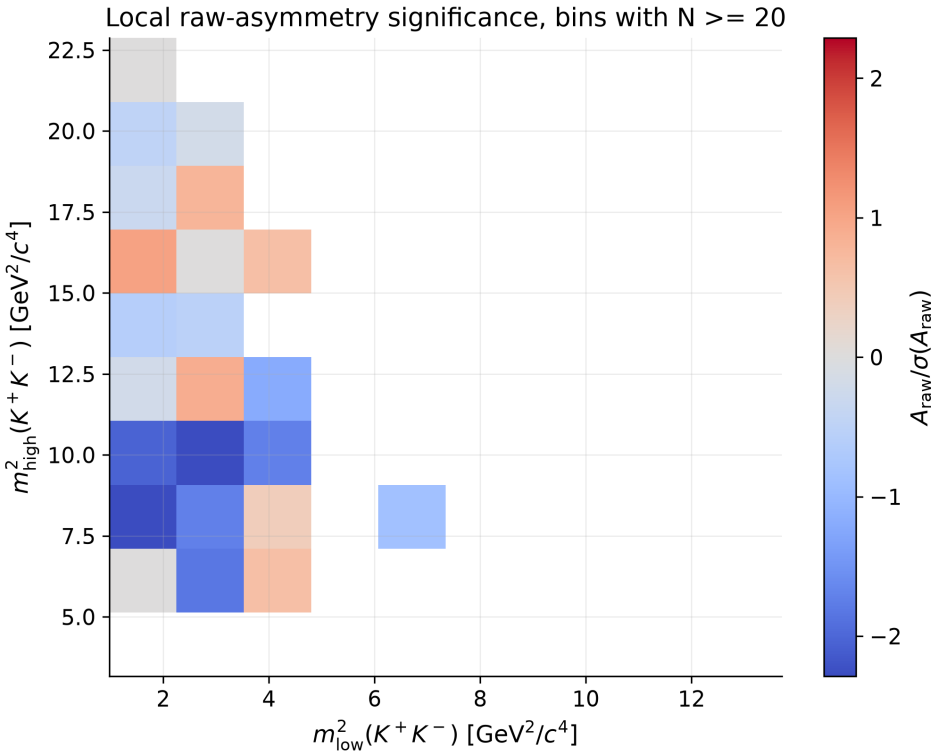


Figure 15: Local raw-asymmetry significance in the charge-ordered $m^2_{\mathrm{low}}(K^+K^-)$ versus $m^2_{\mathrm{high}}(K^+K^-)$ Dalitz plane. Bins with fewer than 20 selected candidates are masked.

A proper local CP-asymmetry study would require signal-yield extraction or background subtraction in each Dalitz region, efficiency and acceptance corrections, treatment of migration between bins, and detector- and production-asymmetry corrections. The maps shown here should therefore be read as a conservative exploratory diagnostic rather than as evidence for a physical CP asymmetry.

9 Summary and Conclusion

This project implements a complete introductory HEP analysis chain on LHCb CERN Open Data. Starting from ROOT ntuples, the analysis reads selected branches with `uproot`, processes candidates with `pandas` and `numpy`, reconstructs invariant masses, applies physics-motivated selections, fits the mass spectrum by binned maximum likelihood, separates the charge-conjugate samples, validates the statistical procedure with pseudo-experiments, evaluates two categories of systematic uncertainty, and examines the three-body phase space with Dalitz-plot distributions and a local asymmetry diagnostic.

Table 3 collects the quantitative results.

Table 3: Summary of the results obtained in this report. All asymmetries are raw and uncorrected.

Quantity	Value
Candidates in the input sample	5 135 823
Candidates after baseline selection	1645
Candidates inside the fitted mass range	1445
Fitted signal yield	1331 ± 39
Fitted background yield	114 ± 17
Fitted peak position	5284.7 ± 0.6 MeV/c^2
Fitted mass resolution	19.5 ± 0.5 MeV/c^2
χ^2/ndof of the nominal fit	0.90
$N(B^+)$	689.1 ± 28.6
$N(B^-)$	642.1 ± 26.2
Toy p-value of the B^+ / B^- fits	0.035 / 0.43
A_{raw}	$-0.0353 \pm 0.0290\,(\text{stat.}) \pm 0.0437\,(\text{syst.})$
Toy-MC pull width / bias	0.9999 / $< 10^{-4}$
Counting asymmetry in the signal window	-0.0416 ± 0.0275
Largest local Dalitz significance (23 bins)	2.3 (global $p = 0.40$)
Classifier AUC (sideband labels)	0.528

The invariant-mass reconstruction shows a clear B-candidate peak after the baseline selection, at a position consistent with the known $B^{\pm}$ mass [11]. The charge-separated fits give a raw asymmetry compatible with zero within the quoted uncertainties, and the pseudo-experiments show that the statistical procedure and its uncertainty are correctly calibrated. Two results of this report are negative ones, and they are stated as such: the sideband Random Forest does not separate signal from background on this sample, and the double-Gaussian signal shape cannot be constrained by it. A third finding qualifies the result itself: the fit-model and fit-range systematics are two symptoms of one unmodelled low-mass shoulder in the B^+ spectrum, so the systematic is taken from the joint envelope of both variations and exceeds the statistical uncertainty.

The analysis remains intentionally simplified and does not constitute a precision flavour-physics measurement. Several improvements would be required for a full experimental analysis:

- model the partially reconstructed and misidentified contributions below 5200 MeV/c^2 explicitly, instead of excluding that region; their tail under the B^+ peak drives the whole systematic budget here;
- combine the MagnetUp and MagnetDown samples to cancel charge-dependent detector effects, and use a control channel to remove production and detection asymmetries;
- include detector effects, PID calibration, trigger efficiencies, and reconstruction asymmetries, validated on simulation and control samples;
- implement signal-shape alternatives such as a Crystal Ball function, on a sample large enough to constrain them;
- train the classifier before the tight preselection, with cross-validation and overtraining checks, if it is to be useful at all;
- perform the local Dalitz asymmetry study with per-bin background subtraction and efficiency corrections.

Despite these limitations, the project reproduces the structure of a real experimental analysis: transparent event reconstruction, documented selection choices, modular code, reproducible figures,

signal extraction by likelihood fit, uncertainty propagation, validation on pseudo-data, explicit and pre-registered acceptance criteria for systematic variations, and a clear separation between what has been measured and what would still be needed to claim a physics result.

References

[1] N. Cabibbo. Unitary Symmetry and Leptonic Decays. *Physical Review Letters*, 10:531–533, 1963.

[2] Makoto Kobayashi and Toshihide Maskawa. CP-Violation in the Renormalizable Theory of Weak Interaction. *Progress of Theoretical Physics*, 49:652–657, 1973.

[3] CERN. CERN Open Data Portal. `https://opendata.cern.ch/`, 2024. Accessed for open high-energy-physics data and documentation.

[4] LHCb Collaboration. The LHCb Detector at the LHC. *Journal of Instrumentation*, 3:S08005, 2008.

[5] LHCb Collaboration. Matter Antimatter Differences (B meson decays to three hadrons) — Data Files. `https://opendata.cern.ch/record/4900`, 2017. DOI: 10.7483/OPENDATA.LHCB.AOF7.JH09, released under CC0 1.0.

[6] Rene Brun and Fons Rademakers. ROOT: An object oriented data analysis framework. *Nuclear Instruments and Methods in Physics Research Section A*, 389(1–2):81–86, 1997.

[7] Charles R. Harris et al. Array programming with NumPy. *Nature*, 585:357–362, 2020.

[8] J. D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007.

[9] Eduardo Rodrigues et al. The Scikit-HEP Project: Overview and Prospects. *EPJ Web of Conferences*, 245:06028, 2020. Software ecosystem providing `uproot` and `iminuit`.

[10] F. James and M. Roos. Minuit: A System for Function Minimization and Analysis of the Parameter Errors and Correlations. *Computer Physics Communications*, 10:343–367, 1975.

[11] S. Navas et al. Review of Particle Physics. *Physical Review D*, 110:030001, 2024. Particle Data Group.

[12] F. Pedregosa et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[13] Steve Baker and Robert D. Cousins. Clarification of the use of chi-square and likelihood functions in fits to histograms. *Nucl. Instrum. Meth.*, 221:437–442, 1984.

[14] LHCb Collaboration. Measurement of CP violation in the phase space of $B^{\pm} \to K^{\pm}\pi^{+}\pi^{-}$ and $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays. *Physical Review Letters*, 111:101801, 2013. arXiv:1306.1246.

[15] Roger Barlow. Systematic errors: facts and fictions. In *Conference on Advanced Statistical Techniques in Particle Physics*, pages 134–144, Durham, 2002.